Default GRU design: filtered retrieval + zero item initialization

Three GRU layers of 32 units | 128-unit prediction head | 215,233 parameters for the illustrated vocabularies

A Encode the inputs available before response t

3 timing values + availability flag
Recency; item age; prior mean spacing
4 numeric coordinates

4 LKT predictors
Intercept; student logitdec;
student-item logitdec; student-item PPES

Current item ID
9,134 known items

Current skill-combination ID
1,607 known combinations

Time encoder
Linear 4 → 32; ReLU
Linear 32 → 16
16 learned time values

History-feature encoder
Linear 4 → 32; ReLU
Linear 32 → 16
16 learned history values

Trainable item embedding: 9,135 × 16
All item vectors START AT ZERO
Then learn 16 values per item
9,134 known + 1 reserved zero row

Skill embedding: 1,608 × 16
Random start; then learn normally
1,607 known + 1 reserved zero row
16 values per skill combination

Concatenate 16 time + 16 history + 16 item + 16 skill = 64 values
Input dropout 0.1 produces x(t), shared by retrieval, prediction and memory update

Recency = time since the previous response (any item). Item age and prior spacing use this student-item exposure history, including studies.

B Filter earlier memories, then retrieve: one head, width 32

Current event e(t): 96 values
Previous h3(t−1): 32 + x(t): 64

Past completed events from this participant
Newest 100 globally: e(j): 96; post-answer h3(j): 32; raw item/skill IDs

FILTER: past item = current item OR past skill = current skill (compare raw IDs)

Shared Q/K linear 96 → 32
Query q(t): 32 values
Uses current e(t)

Shared Q/K linear 96 → 32
Key k(j): 32 values per event
Uses past e(j)

Value linear 32 → 32
v(j): 32 values per event
Uses past post-answer h3(j)

Scores q(t) · k(j) / sqrt(32)
Past 100 AND (same item OR skill)
Softmax over eligible events only
Zero weight on all other events

Weighted sum: Σ a(t,j) v(j), 32 values
Output linear 32 → 32
Retrieved memory r(t): 32 values
No eligible memory gives r(t) = 0

Filter the last 100 global events, not the last 100 matches. Q/K share weights and bias; no attention decay or dropout.

C Predict the current response

Retrieved r(t): 32 + current x(t): 64
Concatenate 96 values

Linear 96 → 128
ReLU
Head dropout 0.1

Linear 128 → 1
One logit

Sigmoid + clipping
p(correct at trial t)
ε = 10^{-5}

The current answer is still unknown. Retrieved memory replaces the direct h3(t−1) input to this prediction head.

D Observe the answer, update all three GRU layers, then write the bank

Observed response y(t)
0 = incorrect; 1 = correct

Response encoder
Linear 1 → 32; ReLU; linear 32 → 16
16 learned answer values

Current x(t): 64 + encoded y(t): 16
80 values into GRU layer 1

Previous h1(t−1)
32 state values

Previous h2(t−1)
32 state values

Previous h3(t−1)
32 state values

GRU layer 1
32 units
h1(t): 32 values

Dropout
0.1

GRU layer 2
32 units
h2(t): 32 values

Dropout
0.1

GRU layer 3
32 units
h3(t): 32 values

Keep all three GRU states for the next trial
h1(t), h2(t), h3(t): 3 × 32 values

For the next trial: append [e(t): 96, new h3(t): 32, raw item/skill IDs]
Keep the newest 100 global events; e(t) is pre-answer, h3(t) is post-answer

All encoders, embeddings, GRU layers, retrieval projections and prediction weights learn jointly from response-prediction loss.

At a new participant: zero GRU states and an empty bank; keep the trained embeddings. Retrieval feeds prediction only.

Timing values use log(1 + seconds) and training-only scaling. Undefined prior spacing stays missing in the data, with an availability flag.

Only completed scored responses enter the bank. Dropout is off during evaluation. States and bank carry forward; gradients detach at 100-event BPTT chunk boundaries.